ADVANCED DATA VISULIZATION PROJECT REPORT

(Project Semester January-May 2026)

Crime & Arrest Insights Dashboard

Submitted by

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DECLARATION

I, Pohap Kumar Layal, student of M.Tech Data Science & Analytics under School of Computing and AI at Lovely Professional University, Punjab, hereby declare that all the information furnished in this project report is based on my own intensive work and is genuine.

Date: Signature

Registration No. 12502458 Pohap Kumar Layal

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CERTIFICATE

This is to certify Pohap Kumar Layal, bearing Registration no. 12502458 has completed Advanced Data Visualization (INT555) project titled, **"Crime & Arrest Insights Dashboard"** under my guidance and supervision. To the best of my knowledge, the present work is the result of his original development, effort and study.

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ACKNOWLEGEMENT

The successful completion of this project would not have been possible without the support, guidance, and encouragement of many individuals. We feel truly privileged to have received their assistance throughout the journey of this work.

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ABSTRACT

This project focuses on the development of an interactive Crime & Arrest Insights Dashboard (2010–2019) using real-world arrest data obtained from Data.gov. The primary objective of this study is to analyze and visualize large-scale arrest data in order to identify meaningful patterns, trends, and insights related to crime across different dimensions such as time, location, and demographics.

The dataset used in this project consists of approximately 1.32 million records and 25 attributes, including details such as arrest date, area name, age, sex, crime type, and geographical coordinates. Due to the large size and complexity of the dataset, extensive data preprocessing steps were performed, including handling missing values, removing irrelevant columns, and transforming raw data into a structured format suitable for analysis. Additional columns such as Year, Month Name, and Hour were created to enable time-based analysis.

The dashboard was developed using Microsoft Power BI, and various DAX (Data Analysis Expressions) measures were implemented to compute key performance indicators (KPIs) such as Total Arrests, Average Age, Male Percentage, and Top Year (year with highest arrests). These KPIs provide a quick overview of the dataset and help in understanding the overall trends.

Several visualizations were designed to support detailed analysis. A line chart was used to examine yearly trends, while a monthly trend chart helped identify seasonal patterns. Bar charts were used to highlight the top crime categories and high-crime areas, enabling comparative analysis. A column chart was used to analyze arrest distribution across different hours of the day, helping identify peak crime periods. A donut chart was used to represent gender distribution, and a map visualization was used to identify geographical crime hotspots using latitude and longitude data.

The analysis revealed several key insights. There is a noticeable declining trend in arrests over the years. Certain areas consistently report higher arrest rates, indicating potential hotspots. The data also shows that arrests are predominantly male-dominated, with approximately 79% of arrests involving males. Additionally, specific hours of the day show higher arrest activity, indicating time-based patterns in crime occurrence.

Overall, this project demonstrates the effective use of data visualization and business intelligence tools to transform large and complex datasets into meaningful and actionable insights. It highlights the importance of data-driven decision making in understanding crime patterns and supports the use of interactive dashboards for analytical exploration.

2. INTRODUCTION

In recent years, the availability of large-scale public datasets has significantly enhanced the ability to analyze and understand real-world problems using data-driven approaches. Crime data analysis is one such important domain where data plays a crucial role in identifying patterns, trends, and anomalies. Law enforcement agencies and policymakers rely on crime data to understand the nature of criminal activities, allocate resources efficiently, and implement preventive measures. With the growth of digital records, analyzing historical crime data has become essential for gaining insights into behavioral and geographical patterns of crime.

Traditionally, crime analysis was performed using manual methods and static reports, which were often time-consuming and limited in scope. However, with the advancement of data analytics and visualization tools, it is now possible to analyze large datasets efficiently and extract meaningful insights in a more interactive and dynamic way. Crime data analysis helps in identifying high-risk areas, understanding time-based patterns such as peak crime hours, and analyzing demographic factors associated with criminal activities. These insights are valuable for improving public safety and supporting informed decision-making.

Data-driven decision making plays a vital role in modern organizations and governance. Instead of relying on intuition or assumptions, decisions are based on factual data and analytical evidence. In the context of crime analysis, this approach helps in designing better policies, optimizing law enforcement strategies, and predicting potential risks. By using data visualization tools, complex datasets can be simplified into intuitive charts and dashboards, making it easier for stakeholders to interpret and act upon the information.

Dashboard analytics is an effective way to present data in a structured and visually appealing manner. Interactive dashboards allow users to explore data through filters, slicers, and drill-down features, enabling deeper analysis without technical complexity. In this project, Microsoft Power BI is used to develop an interactive dashboard that provides a comprehensive view of arrest data from 2010 to 2019. The dashboard integrates multiple visualizations such as line charts, bar charts, maps, and KPI indicators to present insights related to trends, crime distribution, and demographics.

Overall, this project demonstrates how dashboard analytics can be used to transform raw crime data into meaningful insights, supporting data-driven decision making and enhancing the understanding of crime patterns over time.

3. PROBLEM STATEMENT

The increasing availability of large-scale crime and arrest data presents significant opportunities for analysis; however, raw datasets in their original form are often unstructured, complex, and difficult to interpret. The arrest data used in this study contains a large number of records with multiple attributes such as date, location, crime type, and demographic details. Without proper processing and visualization, it becomes challenging to extract meaningful insights from such data.

One of the primary issues is the lack of structured insights from raw arrest data. The dataset does not directly provide summarized information about trends, patterns, or key performance indicators. As a result, users are unable to quickly understand important aspects such as total arrests over time, distribution of crime categories, or demographic characteristics without performing detailed analysis.

Another major challenge is the difficulty in identifying trends, hotspots, and patterns. It is not easy to detect whether arrest rates are increasing or decreasing over the years, which areas experience higher crime rates, or at what times crimes are most likely to occur. The absence of visual tools makes it difficult to recognize these patterns and relationships within the data.

Additionally, the large size of the dataset further complicates analysis, as manual methods or traditional reporting techniques are insufficient to handle such volume efficiently. This leads to a need for an interactive and scalable solution that can process large datasets and present insights in a clear and understandable manner.

Therefore, there is a need to transform raw arrest data into a structured, interactive, and visually intuitive format that enables users to explore trends, identify high-risk areas, analyze crime categories, and understand time-based and demographic patterns effectively.

4. OBJECTIVES OF THE STUDY

The primary objective of this study is to analyze arrest data from 2010 to 2019 and extract meaningful insights using data visualization and analytical techniques. The specific objectives of the study are as follows:

- To analyze yearly and monthly arrest trends in order to understand how arrest patterns have changed over time and to identify any increasing or decreasing trends.
- To identify high-crime areas and geographical hotspots by analyzing location-based data, enabling better understanding of regions with higher arrest frequencies.
- To understand crime category distribution by examining different types of offenses, helping to determine which crime categories contribute most to overall arrests.
- To study time-based patterns of arrests, including hourly distribution and daily variations, in order to identify peak hours and periods of high criminal activity.
- To analyze demographic patterns such as age and gender, providing insights into the characteristics of individuals involved in arrests.

Overall, these objectives aim to transform raw arrest data into structured and actionable insights that can support better analysis, visualization, and decision-making.

5. DATASET DESCRIPTION

5.1 Data Source

The dataset used in this project is sourced from Data.gov, which is an official platform providing open government data for public use. The dataset contains arrest records collected over a period of ten years, from 2010 to 2019. It provides comprehensive information related to arrests, including details about time, location, demographics, and crime types.

Dataset Link:

https://catalog.data.gov/dataset/arrest-data-from-2010-to-2019

5.2 Dataset Size

The dataset is large-scale and consists of approximately 1.32 million records (rows), making it suitable for real-world data analysis. It contains around 25 columns (attributes), each representing different aspects of arrest data. Due to its size, efficient data handling and optimization techniques were required to ensure smooth analysis and visualization in Power BI.

5.3 Features/Attributes

The dataset includes several important attributes that are used for analysis. Some of the key features are:

- Arrest Date: Represents the date on which the arrest was made. This field is crucial for performing time-baed analysis such as yearly and monthly trends.

- Area Name: Indicates the geographical area where the arrest occurred. This helps in identifying high-crime areas and regional patterns.

- Age: Represents the age of the individual at the time of arrest. It is used for demographic analysis and calculating average age.

- Sex Code: Indicates the gender of the individual (e.g., Male or Female). This is used to analyze gender distribution in arrests.

- Charge Group Description: Describes the category of crime associated with the arrest. It is used to analyze the distribution of different crime types.

- Latitude and Longitude: These fields represent the geographical coordinates of the arrest location. They are used for map-based visualizations to identify crime hotspots.

These attributes collectively enable comprehensive analysis across multiple dimensions, including time, location, and demographics, making the dataset suitable for building an interactive analytical dashboard.

6. DATA PREPROCESSING

Data preprocessing is an important step in any data analysis project. Since the dataset used in this project is large and contains raw records, it required cleaning, transformation, and optimization before visualization. These steps helped in improving data quality, ensuring accuracy, and enhancing performance in Power BI.

6.1 Data Cleaning

In the initial stage, the dataset was examined for missing and inconsistent values.

- Records containing null or missing values in important fields were either removed or ignored, depending on their impact on analysis.
- Some columns contained inconsistent or irregular entries, which were standardized to maintain uniformity.
- Data types of columns were verified and corrected where necessary (for example, converting date fields into proper date format).

These steps ensured that the dataset was clean, reliable, and ready for further processing.

6.2 Data Transformation

To enable better analysis, several new columns were created from existing data:

- Year: Extracted from the arrest date to analyze yearly trends.
- Month Name: Created to study monthly patterns and seasonality.
- Hour: Extracted to analyze time-based patterns and identify peak crime hours.

These transformations helped in converting raw data into meaningful formats that could be easily used in visualizations.

6.3 Data Reduction

Since the dataset contained many columns, not all were relevant for analysis.

- Irrelevant and unused columns were removed to simplify the dataset.
- This reduction helped in improving performance and reducing memory usage in Power BI.
- Only the columns necessary for analysis (such as date, area, age, gender, crime type, and location) were retained.

Overall, data preprocessing ensured that the dataset was clean, structured, and optimized, enabling efficient analysis and smooth dashboard performance.

7. DATA MODELING

Data modeling is a crucial step in the development of a Power BI dashboard, as it defines how data is structured, related, and used for analysis. In this project, a simple and efficient data model was created to handle the dataset and support analytical requirements.

7.1 Data Model Structure

The dataset was modeled using a single table approach (fact table). Since all required information such as date, area, age, gender, crime type, and location were available in one dataset, there was no need to create separate dimension tables.

- Each row in the table represents a single arrest record.
- The table contains all relevant attributes required for analysis.

This approach simplified the data model and made it easier to implement for this project.

7.2 Measures Using DAX

To perform calculations and generate insights, DAX (Data Analysis Expressions) was used to create various measures.

Some important measures include:

- Total Arrests: Calculated using COUNT of records

```
Total Arrests = COUNTROWS('Arrest_Data_from_2010_to_2019')
```

- Average Age: Calculated using AVERAGE of the age column
- Male Percentage: Calculated using filtered aggregation

```
Male % = DIVIDE (CALCULATE ([Total Arrests],'Arrest_Data_from_2010_to_2019'[Sex Code] =
"M”), [Total Arrests]) *100
```

- Top Year: Identified using ranking logic

```
Top Year = CALCULATE(MAX('Arrest_Data_from_2010_to_2019'[Year]), TOPN (1,
VALUES('Arrest_Data_from_2010_to_2019'[Year]), [Total Arrests], DESC))
```

DAX measures allow dynamic calculations based on filters and slicers, making the dashboard interactive and responsive.

7.3 Aggregation Techniques

Aggregation was mainly performed using the COUNT function, as each row represents one arrest.

- COUNT was used to calculate total number of arrests
- Aggregations were applied across different dimensions such as:
    - Year
    - Area
    - Crime category
    - Hour

This enabled efficient summarization of data and helped in generating meaningful visual insights.

8. DAX MEASURES

DAX (Data Analysis Expressions) is used in Power BI to create calculated measures that help in analyzing and summarizing data dynamically. In this project, several important measures were created to extract key insights from the dataset.

8.1 Total Arrests

This measure calculates the total number of arrest records present in the dataset.

- It is calculated using the COUNT function, where each row represents one arrest.
- This KPI provides an overall view of the dataset size and is used across all visualizations.

Example logic: Count of all rows in the dataset.

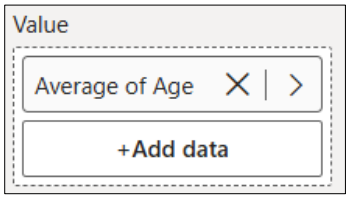
```
Total Arrests = COUNTROWS('Arrest_Data_from_2010_to_2019')
```

8.2 Average Age

This measure calculates the average age of individuals involved in arrests.

- It is computed using the AVERAGE function on the Age column.
- This helps in understanding the general age group involved in criminal activities.

Example logic: Average of the Age column.



8.3 Male Percentage

This measure calculates the percentage of arrests involving male individuals.

- It is computed by dividing the number of male arrests by total arrests and multiplying by 100.
- A filter condition is applied on the Sex Code column to count only male records.

Example logic: (Number of male arrests ÷ Total arrests) × 100

```
Male % =
DIVIDE(
    CALCULATE(
        [Total Arrests],
        'Arrest_Data_from_2010_to_2019'[Sex Code] = "M"
    ),
    [Total Arrests]
)*100
```

8.4 Top Year

This measure identifies the year with the highest number of arrests.

- It uses ranking logic to compare total arrests across different years.
- The year with the maximum value is selected and displayed as a KPI.

Example logic: Find the year with maximum total arrests using aggregation and ranking.

```
Top Year =
CALCULATE(
    MAX('Arrest_Data_from_2010_to_2019'[Year]),
    TOPN(
        1,
        VALUES('Arrest_Data_from_2010_to_2019'[Year]),
        [Total Arrests],
        DESC
    )
)
```

8.5 Peak Hour

This measure identifies the hour of the day during which the highest number of arrests occur.

- It groups data by hour and calculates total arrests for each hour.
- The hour with the highest count is selected.

Example logic: Find the hour with maximum number of arrests.

Summary

These DAX measures play a crucial role in transforming raw data into meaningful metrics. They enable dynamic calculations based on filters and user interaction, making the dashboard more interactive and insightful.

9. DASHBOARD DESIGN

The dashboard was designed to provide a clear, interactive, and user-friendly view of the arrest data. The main focus was to present key insights in a structured manner using KPIs and visualizations. The layout follows a logical flow where users can first view summary metrics and then explore detailed insights through different charts.

9.1 Key Performance Indicators (KPIs)

KPIs are used to provide a quick summary of the most important metrics in the dataset. These are placed at the top of the dashboard for immediate visibility.

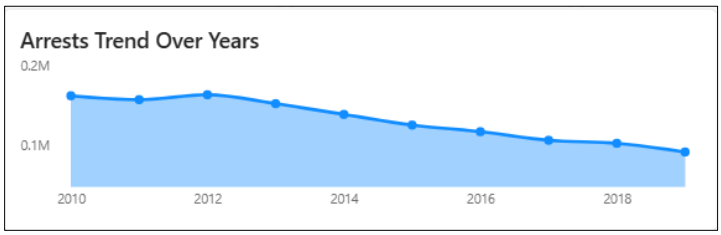
- Total Arrests: Displays the total number of arrest records in the dataset. It gives an overall measure of the dataset size.
- Average Age: Shows the average age of individuals involved in arrests, helping in demographic understanding.
- Male %: Represents the percentage of arrests involving male individuals, highlighting gender distribution.
- Top Year: Identifies the year with the highest number of arrests, providing insight into peak activity periods.

These KPIs allow users to quickly understand the overall scenario before analyzing detailed visuals.

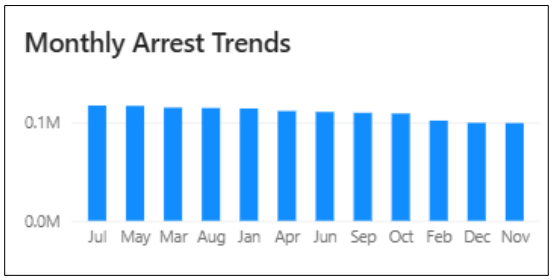
9.2 Visualizations

Various visualizations were used to represent data from different perspectives:

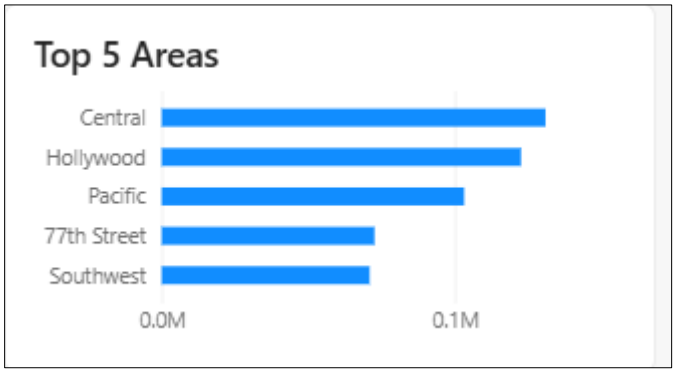
- Line Chart (Year Trend): Used to analyze arrest trends over the years (2010–2019). It helps in identifying whether arrests are increasing or decreasing over time.



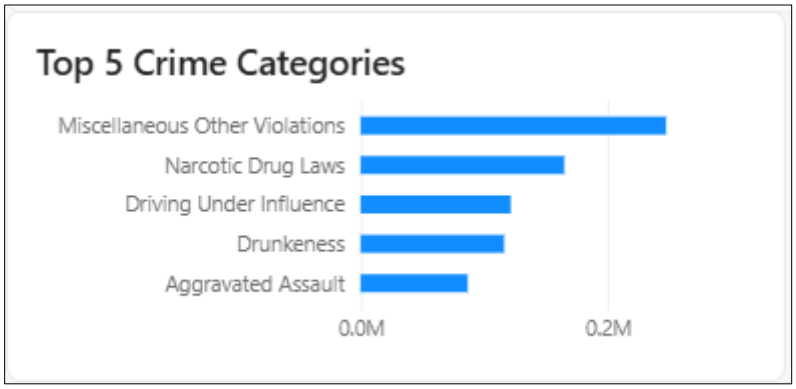
- Column Chart (Monthly Trend): Displays arrests by month to identify seasonal patterns and variations throughout the year.



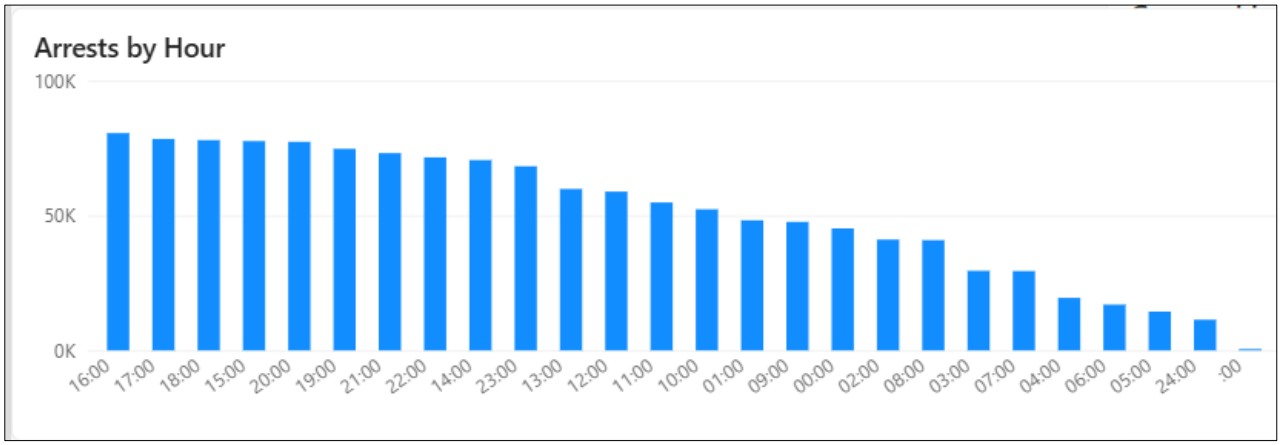
- Bar Chart (Top Areas): Shows the top areas with the highest number of arrests. This helps in identifying high-crime regions.



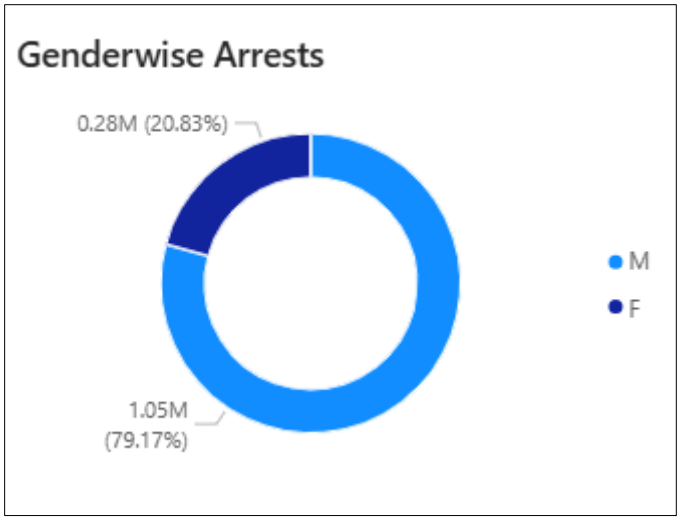
- Bar Chart (Top Crime Categories): Highlights the most common types of crimes based on arrest data, allowing comparison between categories.



- Column Chart (Arrests by Hour): Represents the distribution of arrests across different hours of the day, helping to identify peak crime hours.



- Donut Chart (Gender Distribution): Displays the proportion of arrests by gender, providing a quick demographic overview.



- Map (Crime Hotspots): Uses latitude and longitude data to visually represent arrest locations, helping to identify geographical hotspots.

Summary

The dashboard design ensures that users can easily navigate from high-level insights (KPIs) to detailed analysis (charts and map). The combination of different visualizations allows multi-dimensional analysis of the data, including time, location, and demographics.

10. DASHBOARD LAYOUT & DESIGN

The dashboard layout and design were carefully planned to ensure clarity, usability, and effective communication of insights. A structured approach was followed to organize the visuals in a logical and visually appealing manner.

10.1 Three-Row Structured Layout

The dashboard follows a three-row layout to maintain a clean and organized structure:

- Top Row (KPIs): Displays key performance indicators such as Total Arrests, Average Age, Male %, and Top Year. This row provides a quick summary of the dataset.
- Middle Row (Trend Analysis): Includes line and column charts to analyze yearly and monthly trends. This section helps users understand how arrest patterns change over time.
- Bottom Row (Detailed Insights): Contains charts such as Top Areas, Top Crime Categories, Arrests by Hour, Gender distribution, and Map. This row provides deeper insights into location, crime types, and time patterns.

This structured layout ensures a smooth flow from summary to detailed analysis.

10.2 KPI-First Design Approach

The dashboard follows a KPI-first approach, where the most important metrics are displayed at the top. This allows users to quickly understand the overall situation before exploring detailed visualizations.

- KPIs act as a quick reference point
- They improve decision-making by highlighting key metrics immediately

10.3 Use of Slicers for Interactivity

Slicers were added to make the dashboard interactive and user-friendly.

- Slicers allow users to filter data based on:
  - Year
  - Area Name
  - Gender
- This enables users to:
  - Focus on specific segments
  - Perform customized analysis
  - Explore data dynamically

10.4 Color Consistency and Hierarchy

A consistent color scheme was used across the dashboard to improve readability and visual appeal.

- A primary color (such as blue) is used for most visuals
- Accent colors are used to highlight important elements
- Similar visuals use similar colors to maintain consistency

Visual hierarchy is maintained by:

- Placing important elements (KPIs, main charts) in prominent positions
- Using size and spacing to differentiate between primary and secondary visuals

Summary

The dashboard layout and design ensure that information is presented in a clear, structured, and visually consistent manner. The use of KPIs, organized layout, interactive slicers, and proper color hierarchy enhances the overall user experience and makes the dashboard easy to understand and navigate.

11. KEY INSIGHTS

The analysis of the arrest dataset revealed several important insights related to trends, location, demographics, and time-based patterns. These insights help in understanding the overall behavior of crime over the given time period.

11.1 Declining Trend in Arrests Over Years

The yearly trend analysis shows a gradual decline in the number of arrests from 2010 to 2019.

- This may indicate improvements in law enforcement strategies, policy changes, or variations in reporting practices.
- It highlights how crime-related activities or arrest rates have changed over time.

11.2 High-Crime Areas

The analysis of geographical data reveals that certain areas consistently report higher numbers of arrests.

- These areas can be identified as crime hotspots.
- Such insights are useful for resource allocation and targeted policing strategies.

11.3 Gender Distribution

The demographic analysis shows that a significant majority of arrests involve male individuals (approximately 79%).

- This indicates a strong gender imbalance in arrest data.
- It helps in understanding demographic patterns associated with crime.

11.4 Time-Based Crime Patterns

The hourly analysis reveals that arrests are more frequent during specific hours of the day.

- This indicates the presence of peak crime periods.
- Understanding these patterns can help in planning patrols and preventive measures more effectively.

Summary

These insights demonstrate how data analysis can uncover patterns that are not immediately visible in raw data. By examining trends across time, location, and demographics, the dashboard provides valuable information that can support better understanding and decision-making related to crime analysis.

12. TOOLS & TECHNOLOGIES

This project utilizes modern data visualization and analytical tools to process and analyze large-scale arrest data effectively. The main tools and technologies used are described below:

12.1 Microsoft Power BI

Microsoft Power BI is the primary tool used for building the dashboard and performing data analysis.

- It is a powerful business intelligence (BI) tool used for data visualization and reporting.
- Power BI allows integration of large datasets and provides interactive features such as slicers, filters, and drill-down capabilities.
- In this project, Power BI was used to:
    - Import and clean the dataset
    - Create visualizations such as charts, maps, and KPI cards
    - Design an interactive dashboard layout
    - Enable dynamic filtering and user interaction

Power BI helped in transforming raw data into meaningful and visually appealing insights.

12.2 DAX (Data Analysis Expressions)

DAX is a formula language used in Power BI to create calculated measures and perform data analysis.

- It is used for creating **custom calculations** based on dataset requirements.
- In this project, DAX was used to:
    - Calculate key performance indicators (KPIs) such as Total Arrests, Average Age, Male %, and Top Year
    - Perform aggregations and filtering operations
    - Create dynamic measures that respond to slicers and filters

DAX plays a crucial role in enabling advanced data analysis and improving the interactivity of the dashboard.

Summary

The combination of Power BI and DAX provides a powerful environment for data analysis and visualization. These tools allow efficient handling of large datasets and support the creation of interactive dashboards that deliver actionable insights.

13. LIMITATIONS

Although the project provides useful insights into arrest patterns, there are certain limitations that should be considered while interpreting the results.

13.1 Absence of Population Data

The dataset does not include population information for different areas.

- Due to this, it is not possible to calculate per capita crime rates.
- The analysis is based on absolute arrest counts, which may not fully reflect the actual crime intensity relative to population size.

13.2 Lack of External Validation

The analysis is based solely on the available dataset without incorporating external factors.

- There is no validation using additional data sources such as policy changes, socio-economic conditions, or law enforcement strategies.
- Therefore, trends such as the decline in arrests cannot be conclusively attributed to specific causes.

13.3 Limited Demographic Attributes

The dataset includes only basic demographic information such as age and gender.

- Other important attributes such as income level, education, or occupation are not available.
- This limits the ability to perform deeper socio-economic analysis of crime patterns.

Summary

These limitations highlight that while the dashboard provides valuable insights, the results should be interpreted with caution. Incorporating additional data sources and attributes in future work can help overcome these limitations and provide more comprehensive analysis.

14. FUTURE SCOPE

The current project focuses on descriptive analysis of arrest data; however, there are several opportunities to enhance and extend the analysis in the future.

14.1 Predictive Analysis (Forecasting)

Future improvements can include predictive modeling to forecast arrest trends.

- Techniques such as time series forecasting can be used to predict future crime patterns.
- This would help in proactive decision-making and planning for law enforcement agencies.

14.2 Integration of External Datasets

The analysis can be improved by combining the current dataset with additional external data sources.

- Examples include population data, economic indicators, or regional statistics.
- This would enable more advanced analysis such as per capita crime rates and deeper socio-economic insights.

14.3 Drill-Through and Advanced Interactivity

The dashboard can be enhanced by implementing drill-through features.

- Users would be able to navigate from summary visuals to detailed views.
- This would allow deeper exploration of specific areas, crime types, or time periods.

14.4 Use of Advanced Visualizations

More advanced visualizations can be incorporated to improve data representation.

- Heatmaps can be used to show intensity of crime across time (e.g., hour vs month).
- Other advanced visuals such as decomposition trees or clustering techniques can provide deeper analytical insights.

Summary

These enhancements would transform the dashboard from descriptive analytics to more advanced and predictive analytics, providing deeper insights and improving decision-making capabilities.

15. CONCLUSION

The project successfully demonstrates how large-scale arrest data can be analyzed and transformed into meaningful insights using data visualization techniques. By utilizing Power BI and DAX, the raw dataset was processed, structured, and presented in the form of an interactive dashboard.

The analysis provided several key findings, including a declining trend in arrests over the years, identification of high-crime areas, dominance of certain crime categories, and the presence of time-based patterns such as peak arrest hours. Additionally, demographic insights such as gender distribution and average age helped in understanding the characteristics of individuals involved in arrests.

These insights highlight the importance of data-driven analysis in understanding complex real-world problems like crime. Instead of relying on assumptions, the use of data allows for more accurate interpretation and informed decision-making. The dashboard makes it easier to explore and analyze large datasets through interactive visualizations, enabling users to quickly identify patterns and trends.

Overall, this project emphasizes the value of dashboard analytics in converting raw data into actionable insights. It demonstrates how business intelligence tools can be effectively used to support analysis, improve understanding, and assist in decision-making processes.

16. REFERENCES

The following sources were referred to during the development of this project:

- **Data.gov** – Arrest Data from 2010 to 2019
  Available at: https://catalog.data.gov/dataset/arrest-data-from-2010-to-2019
- **Microsoft Power BI Documentation**
  Available at: https://learn.microsoft.com/en-us/power-bi/
- **DAX (Data Analysis Expressions) Documentation**
  Available at: https://learn.microsoft.com/en-us/dax/